

POWER- ENDURANCE



To succeed on climbs like *Quinsana Plus*, 5.13b, at the New River Gorge, WV, climbers need to be able to execute dozens of powerful moves with little rest. Climbers call this type of fitness "Power-Endurance".



Dan Brayack

CHAPTER 8

"Climbing the tenth grade (5.13+) is not just a question of having sufficient power." — Wolfgang Gullich¹

Strong, powerful muscles and expert technique will only get a climber so far. Eventually a dream route will come along that demands the ability to string together a series of difficult moves in order to claim success. The ultimate routes — those that hold the highest place of honor at each crag and occupy our dreams — all demand sustained, intense effort. This requirement to hold it together through a continuous stretch of climbing is part of a great route's draw; this type of climbing is selective, "separating the wheat from the chaff." Routes like these are the most dramatic, with long, sweeping stretches of steep, improbable stone. Such features require climbers to link long sequences of brutally difficult moves. These routes demand the very best from those who seek to climb them, making them the most sought-after routes, and the most rewarding.

This type of climbing is also unsustainable beyond a very brief stretch of time, and climbers know it. Entering into this realm is akin to playing chicken with the muscles' metabolic systems. Most climbers know that once they cross the threshold, it is a race against mounting fatigue and an incapacitating forearm pump. Fortunately, with training, it is possible to "add time" to the proverbial "pump clock," and turn those dream routes into sends.

This brand of climbing is known as *power-endurance* (PE). It is the ability to perform multiple near-maximal climbing moves without rest (in bouts lasting 30 to 180 seconds). It refers to sequences of moves that are so continuous they afford no opportunities to shake (even quickly) or chalk up. The term itself is somewhat controversial because it sounds like an oxymoron, but it is an accurate way to describe this type of training. "Power" refers to the activation of

fast-twitch muscle fibers, a component essential to this type of training that sets it apart from "regular" endurance training. Less-intense endurance training doesn't activate these fibers, and therefore, doesn't improve their endurance. Climbers rely on these fast-twitch fibers to execute difficult moves, so low-intensity endurance training alone is not sufficient for those intent on performing sequences of sustained, powerful moves. Well-executed PE training will improve the metabolic energy pathways of both slow- and fast-twitch muscle fibers, adding precious time to the pump clock and granting a temporary stay of execution when it comes time to go for the redpoint of that dream route.

MUSCLE FATIGUE

Chapter 5: Base-Fitness explained that fatigue develops within skeletal muscle for various reasons. In the case of climbing, fatigue is caused by a culmination of several critical factors, resulting in the debilitating pump that ends many redpoint attempts. A climber attempting a difficult, pumpy rock climb will encounter strenuous holds that require substantial muscular effort to hold onto, and thus a high percentage of the climber's maximum strength. Such intense contractions will:

1. Require a greater rate of ATP production to power muscle contractions.
2. Rely on fast-twitch fibers (which have reduced capacity for aerobic metabolism).
3. Restrict blood flow to the forearms.

The combination of these factors reduces the muscles' ability to utilize the most-efficient means of providing ATP energy for muscle contractions: mitochondrial aerobic respiration. As a result, muscles will turn to glycolysis, which provides ATP at a faster rate but does

so by burning up glucose faster and producing more metabolic waste. Reduced blood flow prevents resupply of glucose fuel and removal of that waste, further exacerbating this problem. Within minutes, harmful metabolites accumulate in the muscles as local supplies of ATP are depleted. These events interfere with cross-bridge cycling of the actin and myosin proteins that create muscle contractions, preventing the climber from holding on. Climbers can avoid or delay this debilitating cycle through power-endurance training.

All Endurance Is Local

In single-pitch rock climbing, fatigue develops and becomes limiting *locally* much more than *systemically*. Compared to many sports, this is rather unusual. In endurance sports like running and cycling, performance is typically limited by the systemic ability of the entire body to intake and process oxygen for aerobic metabolism to fuel muscle activity. Therefore, traits like VO₂ max and heart and lung function are critical (and potentially limiting) factors.

Climbers are typically limited by the performance of the forearm muscles, whose metabolic demands for fuel and oxygen are trivial compared to what the legs consume while running or cycling. Our cardiovascular system can easily provide the relatively little oxygen needed by our forearms during a climb, even without breathing heavily. Therefore, it isn't necessary or beneficial to improve systemic aerobic fitness in most cases, and time and effort spent on this is just a distraction from more-productive activities (granted, it is an enjoyable distraction for many, including the authors).

On the other hand, the local glycolytic and aerobic energy systems within our forearm muscles *will* be pushed to their limit, and quite likely determine success or failure. Effort spent on improving these systems will be rewarded. So-called cardiovascular exercise (or *cardio*, such as running, cycling, swimming, etc.) may promote recovery between climbing workouts by increasing general blood flow throughout the body. Still, it would be preferable to perform a more climbing-specific exercise such as ARC training, which promotes blood flow directly where it is needed as well as includes skill practice. Therefore, while it is advisable to perform cardio for the purpose of promoting overall health and wellness or to manage weight, don't expect it to improve your climbing directly.

EFFECTS OF ENDURANCE TRAINING

Recall that all three metabolic energy pathways (glycolytic, aerobic, and phosphagenic) are active within the muscle cells at all times; it is their relative contribution to muscle function that changes based on the intensity of the activity. Activities that require greater muscular force will require more ATP, and the muscle will first try to provide as much ATP as possible through aerobic respiration (which is sustainable indefinitely). When the limits of aerobic respiration are reached, the muscle increasingly (but not exclusively) relies on glycolysis, which doesn't require oxygen. On difficult, near-limit rock climbs, this threshold is reached very quickly, and the pump clock begins to tick.

Climbing that requires a substantial contribution from glycolysis is unsustainable, and the onset of fatigue and failure is simply a matter of time. The climber's only hope is to complete the difficult climbing before that time arrives — by reaching a rest or the end of the climb. Fortunately, the climber's ability to sustain climbing of this nature can be improved through power-endurance (PE) training.

PE training involves the use of repeated work *intervals* that simulate very intense stretches of climbing, such as route cruxes or long boulder problems. This approach was first developed by runners in the early 1900s, and has since been adopted by all endurance sports. Climbers have used this same basic method for several decades. PE training stimulates adaptations that improve the muscles' ability to perform repeated, high-intensity contractions. These adaptations primarily improve aerobic respiration within the muscle cells³. Aerobic respiration is always occurring in the muscle, even once the maximum steady state (MSS) is surpassed (as it is during PE climbing). Therefore, promoting mitochondrial performance is still a powerful means of improving climbing endurance because it will increase the aerobic production of ATP and removal of metabolites. As described in **Chapter 5: Base-Fitness**, mitochondrial function (quantity, size, and capacity) can be greatly improved through low-intensity endurance training (such as ARCing), but PE training also promotes mitochondrial function.

PE training improves the muscle cells' ability to tolerate sustained muscle contractions by promoting key enzymes in the cells. Enzymes regulate the cells' energetic processes during activity, while also triggering the development of beneficial infrastructure after

exercise. PE training increases the quantity of these enzymes, which then go on to stimulate beneficial adaptations. PE training promotes enzymes that³:

- Increase the quantity, size, and capacity of mitochondria in the exercised muscle, thus increasing the muscle's overall capacity to use aerobic respiration.
- Improve the mitochondria's capacity to oxidize fatty acids and glucose during exercise, thus increasing the rate of ATP production.
- Shuttle nicotinamide adenine dinucleotide (NADH) into the mitochondria for oxidative phosphorylation (thus producing ATP while removing this otherwise-harmful metabolite).
- Initiate the most-efficient ATP production mechanisms at the onset of exercise rather than waiting for fatigue to occur before "kicking into high gear." Thus, PE training allows the muscle cells to anticipate the requirement for ATP when climbing, a key adaptation that delays fatigue by avoiding the use of less-efficient ATP production methods.

All of these adaptations improve the ability of the aerobic energy pathway to provide ATP for muscle function, and, in effect, raise the MSS, thus delaying the transition to glycolysis and eventual muscle fatigue. Even once the MSS is surpassed, these adaptations continue to benefit performance through improved aerobic respiration (which persists even once glycolysis assumes the leading role).

The adaptations caused by endurance training are specific to the actual muscle and fiber used, which has enormous implications for climbers. First, because each grip position uses different muscles and fibers within the forearm, it is important that the holds used in training match those of the climber's goal routes as closely as possible. This will ensure that the endurance needed for the goal route is actually developed during training. To say it plainly, performing laps on slopers in the gym will not improve the crimp endurance needed for an outdoor project, just as lapping the crag's crimpfest testpiece will do little to prepare a climber for a pinch-and-sloper-dominated competition climb.

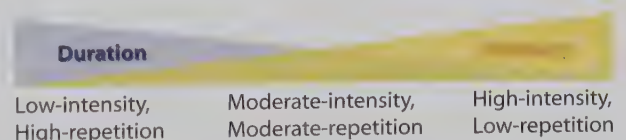
The second consequence of the specificity of endurance-training adaptations is that low-intensity training will not recruit fast-twitch muscle fibers, so low-intensity *endurance* training will not benefit fast-twitch fibers (which are critical to difficult climbing). Therefore, it is necessary for climbers to perform both low- and high-intensity endurance training to ensure

that all types of fibers are trained. Fast-twitch fibers are known to experience a fourfold (or more) increase in mitochondria as a result of endurance training, but only if the training is intense enough to recruit fast-twitch fibers². Such training will, in effect, raise the MSS of the fast-twitch fibers *without reducing their power*, which is enormously beneficial on the most-difficult climbs. The low-intensity ARC training described in [Chapter 5: Base-Fitness](#) primarily benefits Type I (slow-twitch) fibers, while the exercises described in this chapter will provide the correct intensity and duration to promote endurance improvements in the Type IIa and IIb (aka IIx/d) fast-twitch fibers also.

POWER-ENDURANCE TRAINING

Power-endurance training has existed in many forms for decades, at least. The memoirs of legendary climbing heroes contain numerous accounts of punishing bouldering circuits or desperate traverses, repeated beyond the point of exhaustion and frequently revisited as training for their next revolutionary ascent. Still to this day, most climbers agree that interval training is the preferred way to develop power-endurance. Interval training can take many forms, but it generally consists of repeated bouts of climbing effort of a set duration or length, separated by timed rest periods.

The various forms of endurance training create a spectrum that encompasses many training activities, from low-intensity and long-duration sessions (like ARCing), right up to high-intensity, short-duration exercises that border on strength training. The key is to design the training protocol based on strengths, weaknesses, and goals to stimulate the desired improvement. A climber can sculpt a PE routine that lies anywhere on the endurance spectrum (Figure below), but such a routine should be deliberately designed to prepare the climber to accomplish the season's goals.



Endurance Training Spectrum

“One of the best bouldering areas at Stoney is Minus Ten Wall, a thirty-foot-wide wall of vertical limestone, covered in small, polished fingerholds. It was possible to work out endless desperate problems on this, which trained my fingers for the style of climbing on Genesis.” — Jerry Moffat⁵

SPECIFICITY OF ENDURANCE TRAINING

Since adaptations to PE training occur in individual muscle fibers, endurance training is very specific in terms of hold size and type, contraction duration, number of moves (repetitions), and total set duration. Therefore, it is critical to match the training to the climber’s intended goal routes as much as possible. Fortunately, the Power-Endurance Phase directly precedes the Performance Phase, so by the time the PE Phase begins, the goal routes and their characteristics should be very clear. This should remove much of the guesswork from planning your training exercises.

With a specific route (or routes) in mind, the climber should identify the sections of the route that will push him to the limit of his power-endurance. (Some routes may not have this characteristic, and in these cases, PE training may not be a priority for *that route*.) This crux section could be the entire route, but will likely be a stretch of climbing between good rests, spanning at least a few bolts, and often including the route’s hardest moves. Often, such a stretch may be

so demanding that it is not even possible to chalk up in the midst of it, and clipping the bolts may be very difficult. Count the number of movements *per hand* for this crux section and, if possible, estimate how much time it takes to climb it, accounting for the likelihood that, with practice, the sequence can be climbed more quickly. (It’s likely that early in the Performance Phase, it isn’t possible to “link” this entire section, so add up the durations of the links that can be climbed.) There shouldn’t be any good rests in this section, but take note of any poor rests which may be simulated in training. In summary, note all of these physical characteristics of this crux section:

- Number of moves *per hand*
- Handhold size and type
- Foothold size and type
- Wall angle
- Time duration required to climb it (once the moves are rehearsed)
- Appropriate climbing pace
- Rest opportunities

If it’s not possible to physically climb on the goal route (i.e., it’s at a faraway crag), it may be possible to view video of other climbers on it — preferably climbers of similar ability, with similar pacing, who know the best beta! The parameters obtained during this analysis will be used to design specific PE-training exercises.

Endurance adaptations are very specific to the training activity. For best results, train on holds that mimic the size and shape of those on your goal route.



Route Analysis for PE-Training Specificity

Specificity is critical to PE training. When there is a specific goal route in mind, the best approach is to try the route and record its key characteristics. Absent a specific goal, data from a sampling of routes can be used to steer PE training. The best way to collect this data is by analyzing video footage of yourself on past redpoint projects. Here is an example:

Route	Grade	Crag	Hand Moves	Total Time (sec)	Pace (sec/move)	Route Style
Cow Reggae	13b	Wild Iris, WY	18	140	7.78	Steep, big moves, pumpy
Trench Warfare	13d	China Cave, UT	30	200	6.67	Steep, big moves, pumpy
The Heretic	13b	City of Rocks, ID	16	115	7.19	Steep, big moves
Deceivor	13d	Little River Canyon, AL	16	80	5.00	Near-vertical, bouldery, short
Before There Were Nine	13d	Independence Pass, CO	25	160	6.40	Gently overhanging, techy, many complex moves
Busload of Faith	14a	Sinks Canyon, WY	20	114	5.70	Gently overhanging, techy, many complex moves
Calypso	13d	City of Rocks, ID	26	145	5.58	Gently overhanging, techy, many complex moves

This table is meant to describe how to perform the analysis, so don't fixate on the route grades or pace, which will vary between climbers. The pace values are averages, obtained by timing the total duration of the crux section and then dividing it by the number of hand movements. Therefore, any particular move could take much more or less time.

Analyzing your climbing in this way can be very enlightening. First, notice how similar the results are. These are seven very different routes from across the country, and yet they all contain crux sections with between 16 and 26 hand moves, and I climbed them all with a consistent pace; around six seconds per hand move (once the moves were well-rehearsed). Second, it is evident that steeper routes tend to be climbed at a slower pace. This is because they have fewer, but longer moves (such as hip rotations, and large foot movements) that often require multiple foot movements per hand move. On the other hand, technical routes often have many "tic-tac" moves that can be executed quickly, with only minimal body adjustments. This may be surprising because many assume that technical routes are climbed more slowly. They may be climbed slower in terms of "vertical feet per second," but not in terms of "moves per second," which is the more-relevant measure for PE training.

If you haven't done so already, start recording some of your projecting efforts and put the footage to good use, refining your PE training for better performance.

— Mike Anderson

"I think that drive and mental strength are the factors that distinguish ordinary from extraordinary achievements. In competition, it is not the physically strongest climbers who win, but rather the mentally strongest."

— J.B. Tribout⁴

POWER-ENDURANCE TRAINING EXERCISES

Two forms of interval training exercises, each one lying on either end of the endurance spectrum, will be described below. They are *Linked Bouldering Circuits*, and *Route Intervals*. Though both exercises use repeated efforts of sustained climbing, they differ in the adaptations they seek to achieve. Therefore, they require subtle differences in execution, including distinctly different mentalities from the athlete. Linked Bouldering Circuits are used to prepare for routes requiring powerful sprints with little or no rest, where a boulderer's commitment and gymnastic attitude is required. They require higher intensity and arousal.

	Exercise	Simulates	Set Duration (minutes)	Total Hand Moves	Total Sets	Work/Rest Duty Cycle	Rest Between Workouts
High-Intensity	LBC	Long Boulder Problems	0:45 - 1:30	10 - 20	6 - 8	1:2 to 1:1	48 - 72 hrs
Mid-Intensity	LBC	Bouldery "PE" Routes	1:30 - 4:00	20 - 40	4 - 6	1:2 to 1:1	48 - 72 hrs
Low-Intensity	Intervals	Pumpy, "Enduro" Routes	4:00 - 15:00	40 - 100	2 - 4	1:2 to 1:1	48 - 72 hrs

Guidelines for Power-Endurance training.

Route Intervals prepare climbers for long endurance routes with rest points, where concentration, pump management, and precise technical execution are required for success.

Ideally, PE training will be tailored to simulate a specific goal route. With a goal route in mind, select the appropriate type of PE training and use the Table above as a guide to determine the set quantities and rest periods. The driving characteristic should be the duration of the set, because muscle fatigue is more a function of time-under-tension (TUT) than number of repetitions. Therefore, the number of hand movements used in the various types of PE training may vary depending on the typical pace of the climber or the intricacy of the climbing. Analyze the goal route to determine the required duration of effort, then use the table to select the other parameters of the workout. Also, realize that high-intensity moves will generally be climbed considerably faster than moderate moves, which are subsequently faster than low-intensity moves. Therefore, there may be overlap between the various endurance categories (a 90-second high-intensity crux "sprint" may contain more hand movements than a two-minute, mid-intensity "half-sprint," etc.). The *duty cycle* is the ratio of work to rest. Initially, climbers should aim for a duty cycle of 1:2; that is, the set duration should be half the rest duration. As fitness improves, the climber should steadily reduce the work/rest ratio toward a 1:1 duty cycle, in which the rest duration is equal to the set duration.

THE LEGENDARY 4x4

The "4x4" is a prevalent training tool that has many variations, but in its fundamental form consists of a circuit of four carefully selected boulder problems climbed in immediate succession,

followed by a rest period, then climbed again, and so on until the four-boulder-problem circuit has been climbed four times. 4x4s were developed by climbers in Salt Lake City in the early 1990s to train for the short, powerful routes they were developing at the nearby American Fork Canyon. Any climber who has come to grips with the desperates on the Hell Wall will understand why.

4x4s have been a popular and successful training tool for many years, but they have a clear flaw that can easily be improved: the slight rest that is obtained as the climber completes one problem then drops to



4x4s were first developed by Utah climbers to prepare for the short but sustained routes at American Fork Canyon. JC Hunter's *Fantasy Island*, 5.14b, is a perfect example.  Andrew Burr

the ground and moves to the start of the next problem. This does not accurately simulate climbing and violates the imperative for specificity in endurance training. Extremely taxing, dynamic, powerful climbing, like that which often composes a route's crux, will rarely afford much more than a half-second of rest at a time as hands dart from one hold to the next. Therefore, the five or more seconds that pass as the climber drops from the end of one problem to start the next is a virtual eternity to battered forearms gasping for respite. These unrealistic rests reduce training stress and interfere with the desired adaption.

LINKED BOULDERING CIRCUITS

The first form of PE training described, the Linked Bouldering Circuit (LBC), is a form of high- and mid-intensity PE training that will improve the climber's ability to sustain a series of difficult moves. Based on the classic 4x4, a linked bouldering circuit is an unbroken chain of boulder problems or a long boulder traverse consisting of a premeditated number of moves, designed to mimic the goal route as closely as possible. If there isn't a specific goal route in mind, a reasonable target for mid-range PE training is about

24 total hand movements (about 12 per hand), with no rest stances or shakes — chalking up should not be possible. This type of training is designed for routes requiring about 20 uninterrupted high-intensity hand movements (like many of the "hard" routes in the US). The moves should be powerful and dynamic, requiring a boulderer's "go for it" mindset to complete. Thus the climber will be forced to conjure up this attitude in the face of debilitating fatigue, and will learn that such moves can be done even when pumped. This "attitude rehearsal" will transfer nicely to the goal route during the redpointing process.

A single LBC set will take one to four minutes to complete (averaging about six seconds per move), which is generally the time limit of the glycolytic energy pathway. Therefore, assuming the moves are the correct difficulty (more on this to come), the circuit will bring the climber to the limit of glycolysis, stimulating the release of the desired enzymes that trigger beneficial adaptations.

The circuit should be difficult but not desperate; easy enough that the climber can complete it most of the time when she is warmed up but not yet fatigued. It shouldn't have any "stopper" moves, but the



Linked Bouldering Circuits can be created in the gym by linking together existing boulder problems (often with some down-climbing in between). This LBC includes increasingly difficult problems, to train for a pumpy route with a high crux

difficulty can vary as needed to mimic a goal route. When in doubt, err on the side of making the circuit too easy. It's better to complete the sets with energy left over and add more sets at the end than to fall off too early, before sufficient fatigue is developed (which would yield little training benefit). Because of the way fatigue develops in the forearms, an appropriate LBC will not cause a debilitating pump during the first set or two. It may require three or four sets for the pump to develop. While a dizzying pump may not exist, the muscles are still fatiguing, resulting in a loss of power and coordination (which could cause failure without a deep pump). This is the necessary training stimulus to achieve the desired adaptation — the ability to sustain powerful, precise movement for 20 to 30 moves. This is consistent with the type of stress imposed by many limit redpoint routes.

CREATING LINKED BOULDERING CIRCUITS

Building a proper LBC is not trivial and requires some thought and experimentation, which is why the 4x4 framework is helpful. By building up the circuit from three, four, or five distinct boulder problems, it is easy to tweak the composition and difficulty of the overall circuit by manipulating characteristics such as the order in which the problems are climbed, or by substituting problems. For example, consider the three very different circuits constructed from the same pool of five boulder problems rated V2, V2, V3, V5 and V6:

<u>Circuit No. 1</u>	<u>Circuit No. 2</u>	<u>Circuit No. 3</u>
V5	V5	V5
V3	V2	V2
V2	V2	V3
V2	V3	V6

There are dozens of ways this circuit could be constructed, but these were chosen to illustrate a few key points. Assuming each problem is only climbed once, Circuit No. 1 is the easiest possible arrangement because the harder climbing is done early in the circuit, when the climber is less fatigued. By the same token, Circuit No. 3 is quite difficult because the hardest problem comes at the end, when the climber will be much more fatigued. Circuit No. 2 is slightly harder than Circuit No. 1, which was accomplished simply by swapping the order of two problems. This is a simple adjustment that can be used to fine-tune the difficulty of a circuit, or to make the training progressive throughout the season. Ideally, the order of difficulty will *eventually* mimic the goal route.

A further benefit of this circuit-construction method is that it allows the climber to select the appropriate style of problems (angle, hold type, etc.), and keep them constant. If all of the original five problems are of the desired style, rearranging or substituting them still creates a circuit of the desired style. These problems can be worked and rehearsed during the Limit Bouldering routine sessions of the Power Phase (as part of the Warm-Up Boulder Ladder or Hard Bouldering phases of those workouts). Thus, when it is time to include them in LBCs, they will be well-known problems that can be climbed on demand, with a low chance of failure.

ROUTE INTERVALS

Route Intervals are a form of lower-intensity PE training that are usually accomplished by climbing a route or bouldering circuit repeatedly (also known as “route lapping”). Route Intervals are appropriate training for goal routes that are long and sustained, with less-distinct cruxes. Such routes will often include poor rests that provide enough recovery to allow climbing to continue beyond the two-to-three minute time limit that constrains “pure” PE climbing. This does not necessarily mean there are no hard crux moves, only that there are ample rests to allow the effort to continue beyond the typical PE limit.

A benefit of route lapping is that it can be done on lead, thus providing an opportunity to practice all of the challenging components inherent to lead climbing. Route Intervals also make better training for onsight climbing, because the lower-intensity and longer-duration efforts are more specific to onsighting. Therefore, it is common to begin the PE Phase with Linked Bouldering Circuits, then transition to Route Intervals later in the phase (especially when the climber is training for several goal routes with a variety of endurance demands). This coincides with the general decline in power that occurs through the Performance Phase.

Route Intervals can be performed on outdoor routes, indoor routes, or indoor traverse circuits, but route selection is critical. The difficulty should be about one letter grade above the climber's consistent onsight level. Therefore, the climber's goal route usually should not be used for this type of training. Instead, select a route that is less difficult than the goal route, but similar in terms of length, angle, hold type, etc. (refer to the previous section, “Specificity of Endurance Training,” for more details). If practical, it is beneficial and not uncommon to use an outdoor

Route-Setting Considerations for PE Training

Specificity: PE Training is highly specific, so select or create routes that utilize similar hold types, steepness, and style as the goal routes you are training for.

Venues: PE Training circuits can be set in many locations, but some are better than others. A rotating climbing wall (aka a Treadwall®) is the ideal artificial tool for PE training. Problems can be set on these machines that cover one complete revolution of the tread, and start and finish at the same point so the climber can climb continuously from one problem into the next without interruption. Further, some models allow the angle and rotation speed to be adjusted to simulate specific terrain. The bouldering area of a quality climbing gym is great for LBCs (especially if you have route-setting privileges), while the lead wall is preferred for Route Intervals. Avoid lead routes for LBCs, so clipping isn't required. If route-setting privileges aren't granted, the gym staff may be willing to set problems specifically for training. It's always possible to create "unofficial" problems using existing holds or a systems wall, Campus Board, etc. A home bouldering wall is another great option for creating linked circuits that can be used for LBCs or Route Intervals.

Continuity: On bouldering walls, it is important to build a continuous linked circuit. That means downclimbing, and possibly traversing between problems. Use problems that are nearby, and avoid overly easy downclimbs that afford too much rest. Going down should be just as hard as going up. Alternatively, it may be easier to build a traversing circuit, but this type of climbing is less specific, so incorporate alternating sections of up- and downclimbing (not just horizontal movements) into the "traverse."

Rests: High- and mid-intensity PE training should not include resting or shaking within a set, and moves should be so difficult that chalking up isn't feasible. However, low-intensity sets may include chalking or resting, especially if it is possible to mimic a taxing rest position on the goal route. A good way to limit rest opportunities is to eliminate holds that allow matching (and avoid placing two holds side by side). To allow limited rest, set one jug for each hand, separated by a few moves, thus allowing one hand to shake briefly and chalk, but preventing alternating shakes with both hands. The number and duration of rests will impact the total duration of the set, which could push the set duration beyond ten minutes in the case of a long Route Interval.

Pacing: Beware that routes or circuits climbed in the gym will generally be climbed much faster than outdoor routes because the holds are easy to spot and have uniform surfaces, thus less time is required for analysis and precise movement. Climbing too fast can sabotage PE-training effects. As described earlier, time the duration of the goal route's crux section, or gather general data from several routes to guide future PE training (this is also a helpful analysis tool when projecting routes). It may be helpful to create technical, complex movements, with insecure footholds, to slow the climbing pace during PE training. Using pockets or other complex holds for footholds is one proven method for reducing climbing speed. Another method is to establish "time gates" at key holds; do not proceed past these gates until the specified amount of time has passed.

Adjustments: It is unlikely that you'll create your circuit perfectly the first time. Adjust holds and sequences as necessary to achieve the desired level of intensity, length, and level of pump. Be prepared to make changes throughout the season to maintain proper duration and intensity as the sequence becomes "dialed" and PE improves.



A rotating climbing wall is the ultimate tool for PE training.

route at the same crag. Such a route will mimic the goal route better than any gym route possibly could. The training route should be adequately rehearsed so that the climber can “send at will” when she is unfatigued. Therefore, a former project at the same crag that is similar to the current goal route would be ideal for Route Interval training. Strenuous rest and chalk opportunities should be included, but avoid “camping out” at easy stances.

Once the route is selected and adequately rehearsed, a Route Interval workout can be performed. The route will be climbed two to four times in succession, with a duty cycle of roughly 1:2 (time required to complete one lap is half the duration of the rest period), which should be reduced to 1:1 as the climber progresses. Therefore, time the duration of each lap and rest period.

As with all PE training, this is not a race; climb at the appropriate pace for the goal route, including pausing to rest and shake. While LBC training demands a boulderer’s “go for it” attitude, Route Intervals should utilize the controlled, confident attitude of a redpoint climber. The pace should be slower than LBC training, especially if rests are involved. Concentrate on technical details such as breathing, relaxation, grip control, and precise movement. Laps should be climbed on lead if possible, in order to practice the unique elements of lead climbing such as clipping from the best stances, staying relaxed while clipping, climbing confidently through a runout, etc. Route Intervals can be performed on a bouldering traverse (which is easier to set and modify), but doing so sacrifices the benefits of lead-climbing rehearsal and upward movement.

It is common to wrap up a day of limit redpoint climbing with a Route Interval session. This is appropriate when the previous climbing for the day consisted of primarily working crux moves on goal routes (roped bouldering) and did not include taxing, sustained endurance efforts. This is common early in the redpointing process, when just beginning to learn the moves on the project and long “linkage” isn’t yet possible. Beware that by tacking on this extra workload, more rest may be required before the next climbing day. Therefore, it is more advisable to perform these at the end of a climbing weekend. Nevertheless, this is a good way to incorporate some PE training into an otherwise-crowded training schedule. Follow the aforementioned guidelines, but only perform two or three laps at the most. Route Interval training is very similar to route projecting, so it can hasten “burnout” and result in accumulating fatigue that impacts overall

climbing performance. Exercise restraint in performing this type of training.

PERFORMING THE WORKOUT

Once a training route or LBC is prepared, it should be rehearsed sufficiently that failure during training will result from muscle fatigue, not a botched sequence. If training in a public gym, aim for a low-traffic time slot to stay out of the way of other climbers. Each “lap” of the circuit or route is considered one “set.” The duration of each set should be timed and recorded, but this isn’t a race and the goal isn’t to “speed up” each successive repetition. The goal should be to complete the circuit at the appropriate pace (simulating an outdoor route-climbing pace).

If the circuit is the right difficulty, simply completing it once will be quite difficult. As mentioned earlier, rest between sets will decrease from workout to workout as the climber progresses. Early in the PE Phase, aim for a duty cycle of 1:2. As fitness improves, reduce the work/rest ratio toward a 1:1 duty cycle.



Have a partner time your work intervals and rest periods to ensure progression through the PE Phase.

LINKED BOULDERING CIRCUIT (LBC) ROUTINE

WARM-UP PHASE 1:

- **Low-intensity ARC traversing (15 to 20 minutes)**
- Light, active stretching on the wall

WARM-UP PHASE 2:

- **Warm-Up Boulder Ladder (15 minutes)**
- Begin with "V-easy"
- 1 to 3 problems of each grade up to flash level
- No more than three attempts per problem
- Rehearse any problems incorporated in the LBC

WARM-UP PHASE 3:

- **Hard Bouldering (20 minutes)**
- 2 to 3 "realistic" problems — if necessary, rehearse problems that will be used in the LBC
- 3 to 4 quality attempts per problem
- 2 to 5 minutes rest between attempts

LINKED BOULDERING CIRCUIT:

- **Move Selection:** Powerful, dynamic moves that are specific to goal routes and are well-rehearsed, so the circuit can be completed most of the time when un-fatigued.
- **Duty Cycle:** 1:2 early in the phase, progressing to 1:1 late in the phase. (For example, begin the PE Phase by resting four minutes for a two-minute set, progressing toward two minutes' rest for a two-minute set.)
- **Sets:** 6 to 8 for sets of 20 moves or less; 4 to 6 for sets of 20 to 40 moves.
- **Pace:** It's not a race; climb at a pace similar to the goal route and focus on proper breathing, relaxation, grip control, pump management, and precision movement. Time the lap and record it.
- **Notes:**
 - If you fall but aren't too pumped, resume the circuit.
 - If you fall due to fatigue, end the set and note the time and number of hand movements completed.
 - During rest periods, focus on proper breathing and prepare for the next set.
 - Once you cannot complete 75 percent of a set due to fatigue, end the workout.

COOLDOWN:

- Low-intensity ARC traversing (15-20 minutes)
- Rest 10 to 20 minutes after Phase 4 before cooldown (stretch and massage forearms)
- Rehearse techniques
- Light, active stretching on the wall

Anywhere from two to eight sets may be accomplished in a given workout, depending on the climber's goals (which should determine the intensity). Typically, three to six sets are performed, and the last couple of sets will often not be completed successfully due to accumulating fatigue. If all sets are completed, continue the rest and work cycle until fatigue is reached, and then reduce the rest period accordingly for the next workout. There is little value in attempting more sets once fatigue has set in. If it isn't possible to complete at least 75 percent of a given set, end the interval workout and move on to the cooldown. See Table on previous page for more details on rest periods and numbers of sets.

In reality, it is not so easy to anticipate when fatigue will cause failure in the midst of a training session — or a season. Nevertheless, the goal is to create a circuit or route that imparts a massive, building pump and requires considerable effort to complete. Throughout the workout, it should be increasingly difficult to complete each lap, with failure almost certain on the last couple of sets. It is not necessary to fail at the end of every set. It is perfectly acceptable to complete a lap “with a little gas left in the tank,” albeit this process should be hard and accompanied by a pump (high-intensity PE training may not last long enough to develop a debilitating pump). If a fall occurs midway through a set due to some factor other than severe fatigue, jump back on the wall and attempt to complete the set. If a fall occurs as a result of fatigue, end the set.

After each set, record its duration and note if the set was completed or where failure occurred. Focus on good breathing and recovery during rest periods. Sip water and/or pace slowly to aid recovery. Chalk up and be in position to begin the next set when the rest period ends. These are among the most physically exhausting work-

outs a climber will perform. Be prepared to suffer and produce tremendous effort during the last set or two.

While climbers are urged to perform PE training on rock or an artificial climbing wall, it can be done on a more compact apparatus such as a system wall, Campus Board, or even a hangboard. Follow the same protocols described above, and take extra care to ensure that the proper intensity and pace of “climbing” are used. For the Campus Board and hangboard, which would normally be used without feet, add a *kick plate* (a small panel plastered with footholds — see photo in last column) so that the hands can be weighted with realistic loads and you can simulate climbing movement. The hangboard can overhang the kick plate to simulate overhanging climbing.

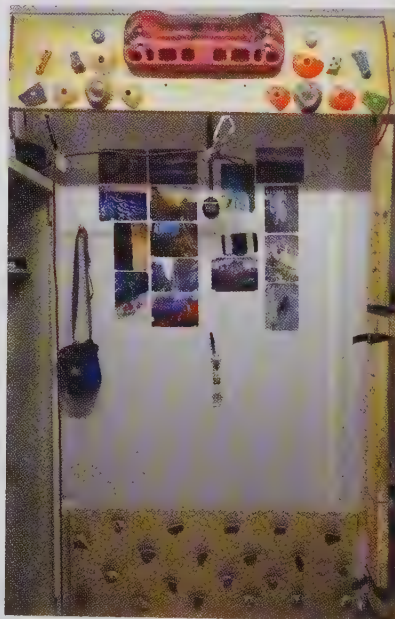
FOCUS AND INTENSITY

During strength and power training, very high arousal is desired to produce maximum force and power. This high state of arousal and accompanying intensity level would lead to premature fatigue in PE training. Therefore, a lower level of arousal should be used, similar to that of a limit redpoint or onsight attempt. The intensity level must be high enough to complete the individual moves of the circuit, but no higher; thus it might be necessary for the climber to adjust arousal and intensity within a circuit. Though challenging, adjusting arousal on the fly is an important skill that is essential for difficult redpoints, and practicing it is one of the key benefits of PE training.

PE circuits are more similar to real-life redpoint attempts than any other training activity in the Rock Prodigy program. Therefore, treat them as rehearsal, practicing the desired behaviors required for performance efforts. Focus on climbing efficiently when *possible*, and turning up the intensity and effort when *necessary* to complete the moves. Use a relaxed grip when possible and focus on executing flawless technique. Maintain steady breathing and stay calm and relaxed. When difficult moves interrupt breathing, practice “Finding Calm” (See [Chapter 3: Skill Development](#)). For suggestions on good performance-climbing behaviors, see [Chapter 13: Redpoint and Onsight Climbing](#).

COOLDOWN

A cooldown is critical after PE training, because such training leaves the muscles packed with harmful metabolites. Moderate cyclic contractions are known to aid blood flow through the muscles, which helps to replenish fuel and remove the waste products of muscle



With a “kick plate”, a campus or hangboard can be used for PE training in a pinch.

ROUTE INTERVAL ROUTINE

INDOOR WARM-UP:

Phase 1: ARC traversing (15 to 30 minutes), or three to four warm-up routes

- Rehearse techniques
- Light, active stretching on the wall

Phase 2: Warm-Up Boulder Ladder or one to two routes (15 minutes)

- Begin with "V-easy"
- 1 to 3 problems of each grade up to flash level
- No more than three attempts per problem
- If route climbing, use routes up to flash level

OUTDOOR WARM-UP*:

Phase 1: Outdoor Mileage routes (two to three moderate routes)

- Rehearse techniques
- Light, active stretching on the rock

Phase 2: Outdoor route (15 minutes)

- Climb one more difficult route: either an onsight one to two letter grades below your OS limit, or a well-rehearsed route slightly above your OS limit — such as the interval training route
- Rest 30 to 45 minutes

** NOTE: If performing Route Intervals at the end of an outdoor limit-redpointing day, skip the Phase 1 and Phase 2 warm-up activities and begin with Phase 3: Route Intervals.*

ROUTE INTERVALS:

- **Route Selection:**
 - One letter grade harder than your onsight level
 - Similar to the goal route in terms of length, angle, hold type, etc.
 - Well-rehearsed and can usually be climbed on demand when un-fatigued
- **Pace:** Climb at a pace similar to the goal route, including rests that would normally be used to redpoint the route. Time the lap and record it.
- **Duty Cycle:** 1:2 early in the phase, progressing to 1:1 late in the phase. (For example, begin the PE Phase by resting 16 minutes for an eight-minute set, progressing toward eight minutes' rest for an eight-minute set.)
- **Sets:** Complete two to four laps

COOLDOWN:

ARC traversing (15 to 30 minutes) or Outdoor Mileage (one to two routes)

- Rest 10 to 20 minutes after Intervals before cooldown (stretch and massage forearms)
- Rehearse techniques
- Light, active stretching on the wall

fatigue. Therefore, after a sufficient rest period of 10 to 20 minutes, each PE training session should finish with at least 20 minutes of ARC climbing (30 minutes is even better). Climbing moderate routes is also acceptable, but ensure that the continuous climbing time is at least 20 minutes, and avoid excessive difficulty (it may be necessary to climb back-to-back routes before swapping with the belayer). If excessive skin wear prevents ARCing, consider light upper-body circuit training as a cooldown.

PROGRESSION

Progression in PE training should not consist of performing the circuit more quickly — that would be regression, as the body is asked to perform shorter sets. In other sports, endurance progression would be indicated by performing the effort for a longer duration or at a faster pace, but neither of these make sense for climbing, in which the goal is to climb a fixed number of difficult moves. Therefore, in climbing, given that a specific goal route is being simulated, it is better to fix the duration and number of moves in the set, and progress in one of two ways (or both). The first option, and the easiest to implement, is to climb the same LBC or Route Interval and progressively reduce the rest period between sets. The second option is to progressively increase the difficulty of the moves, still aiming to complete the circuit before fatigue sets in.

A combination of these approaches is ideal. After analyzing the goal route and building a suitable circuit or training route, progression throughout the PE phase should consist of reducing the rest period and/or tweaking the circuit appropriately to increase the difficulty of each subsequent training session. Increase intensity by swapping out larger holds for smaller holds, adjusting rest points, or reordering the sequence of linked boulder problems as previously described.

SKILL DEVELOPMENT

As described in [Chapter 3: Skill Development](#) and [Chapter 5: Base-Fitness](#), skill development and endurance training go hand in hand. Just as novel movements were learned during the Base-Fitness Phase, those same moves can be further refined and stress-proofed during the PE Phase. However, it requires deliberate attention to do so. A few specific skill-devel-

opment goals, which can be emphasized throughout the training program, should be set at the beginning of the season. By the time the PE phase is reached, these skills should be mature enough to include in LBCs and Route Intervals. If necessary, design the circuit or route to include opportunities to attempt these novel moves.

While performing PE sets, pay attention to these novel moves, but always keep in mind the other climbing fundamentals: proper breathing, relaxation, grip control, pump management, and precision movement. The nature of this training often compels climbers to “race” through the set, but it is not a race. Climb at a reasonable pace, identical to that used on similar redpoints (use time gates, if necessary). Relax, breathe, *think*, and maintain controlled, precise movement. Do not flail! That massive pump will set in whether you’re climbing slow, fast, controlled, or flailing, so climb the right way and make it a habit that will transfer to redpoint attempts — when it really matters.



PE workouts are ideal for stress-proofing novel moves and skills.

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
*WBL 60 min *Linked Bouldering Circ (LBC), DC=2:4	*Skin Care		*WBL 50 min *LBC, Duty Cycle (DC) = 2:3, SE	*Skin Care		*Redpoint (RP) attempts on bouldery routes
*Redpoint (RP) attempts on bouldery routes	*Skin Care		*WBL 50 min *LBC, DC = 2:2 *SE	*Skin Care		*RP Attempts *OM, 2 mod pitches
*RP Attempts *OM, 2 mod pitches	*Skin Care		*WBL 40 min *Intervals, DC = 4:4 *SE	*Skin Care		*RP Attempts on pump routes *OM, 2 mod pitches

Key:

ARC – Aerobic Restoration and Capillarity Training
OM – Outdoor Mileage
OAE – Optional Aerobic Exercise

LBC – Linked Bouldering Circuits
min – Minutes
mod – Moderate
RP – Redpoint

SE – Supplemental Exercises
DC – Duty Cycle (work-to-rest ratio)
WBL – Warm-Up Boulder Ladder

Sample PE training schedule integrated with outdoor climbing. This schedule assumes that Saturday and Sunday are the typical outdoor climbing days. If “three-day weekends” are possible, this can be modified to add a rest day between the RP days. Note: Duty Cycle (DC) is given in minutes (i.e., 2:4 means about 2 minutes of climbing followed by 4 minutes of rest).

“The difference between getting to the last move and doing the last move could be a month of training.”

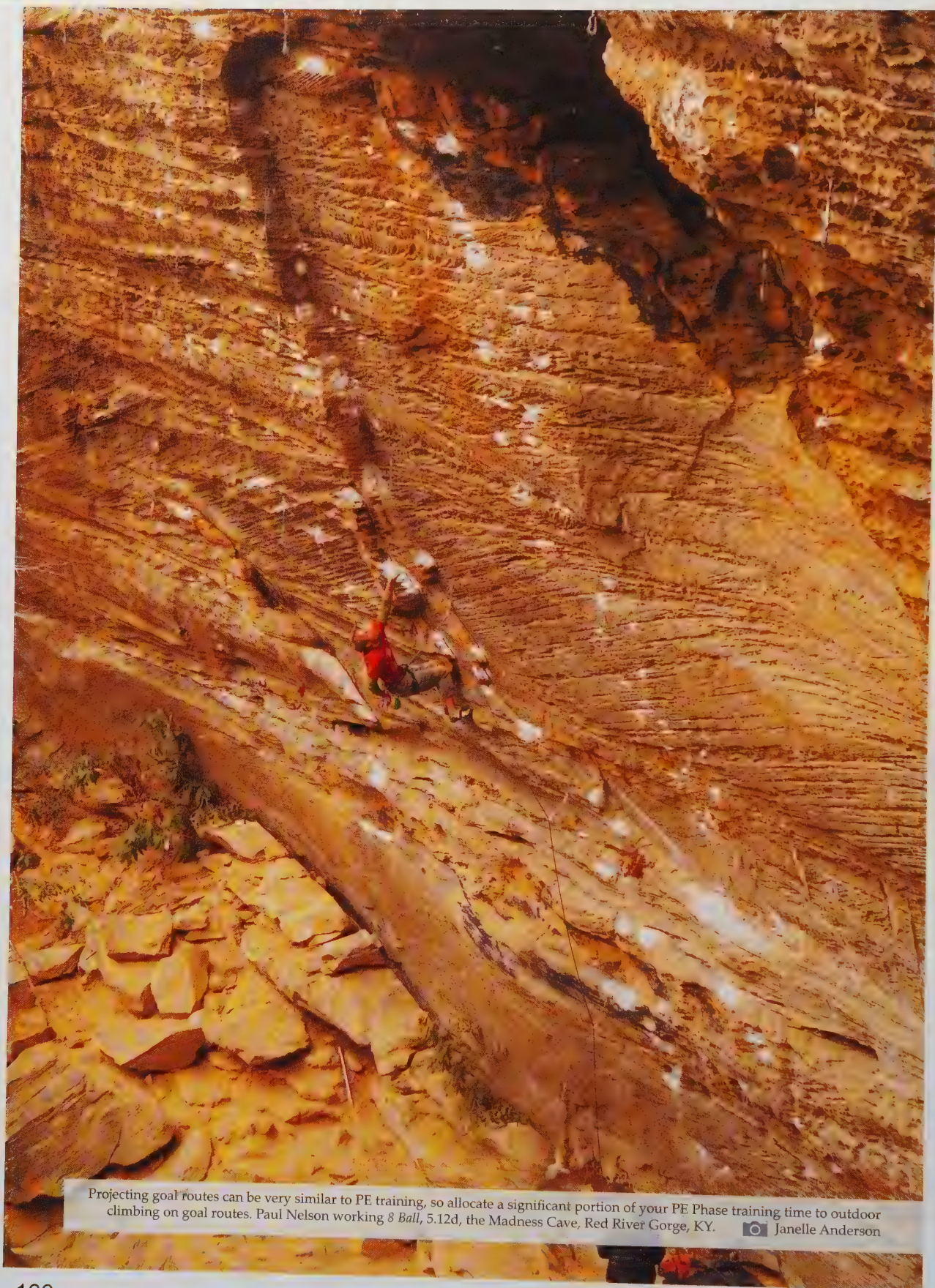
— Jerry Moffat⁵

SCHEDULING

PE training is performed near the end of the macrocycle, beginning immediately following the Power Phase, and continuing through the Performance Phase. Both LBCs and Route Intervals can and should be performed as part of a “healthy diet” of PE training. These are complementary exercises that will benefit different aspects of climbing performance. The importance of PE training exercises in any given season varies greatly, depending on the season’s climbing goals. Very bouldery routes with ample rest might not require any PE training at all, while long, sustained routes without distinct cruxes, or with ample rests, may require only Route Intervals but no LBC training. These factors should be considered when scheduling PE training.

The muscles’ enzymatic adaptations to PE training occur quite rapidly (within a few days), but they are short-lived, lasting only a few weeks. Therefore, training this way early in a season, well before the Performance Phase, accomplishes little and also interferes with more-beneficial activities. Furthermore, continuing PE training beyond about four weeks will likewise accomplish little. The muscles adapt quickly to this training, to a point, after which further training adds little new capability while extending the subsequent recovery time.

PE training should be started one to two weeks prior to the Performance Phase and continue for up to four weeks, into the Performance Phase. The process of projecting and onsighting goal routes is very similar to PE training, and will have similar training effects, so it is not necessary to perform a large volume of dedicated PE training if regular access to goal routes is possible. In the midst of the Performance Phase, if cragging two days per week, climbers should perform no more than one supplemental PE training session per week. As mentioned earlier, a modest volume of Route Intervals can be performed at the end of performance days at the crag by performing two to three laps on a familiar route or by attempting one or two sub-maximal onsights. Following a PE workout, 48 to 72 hours of rest will be required. An example three-week schedule for a climber incorporating route climbing with PE training is shown at the top of the page.



Projecting goal routes can be very similar to PE training, so allocate a significant portion of your PE Phase training time to outdoor climbing on goal routes. Paul Nelson working *8 Ball*, 5.12d, the Madness Cave, Red River Gorge, KY. 📷 Janelle Anderson

Projecting Fitness

The process of working a project route has PE-training benefits, so it's possible to use the projecting process to make endurance gains without any supplemental PE training. However, like any endeavor, the devil is in the details. The typical hangdogging process for learning and rehearsing moves is more akin to bouldering than endurance training, especially at first, when the moves are not yet familiar. However, PE training becomes possible as "linkage" improves. Once you're able to execute the individual moves regularly, try to link multiple moves together. Use the rope or fixed protection to "pull through" stopper moves if necessary. When falls occur, instead of resting, pull back on the wall immediately and continue climbing. Aim to perform a "set" of moves comparable to a LBC in terms of duration and number of moves. Once a set's worth of moves has been climbed, take the appropriate rest, then repeat for the desired number of sets. Used this way, two to three "working burns" can account for six to eight sets of PE training, and a very productive workout in terms of PE training and route rehearsal.

In the winter of 2009, I trained diligently for a trip to Smith Rock, but a month before going I severely sprained my ankle playing basketball (another good reason to limit extraneous cardio). I continued to train on the hangboard, and even did some ARCing, but hard bouldering was out of the question (due to the risk of ground falls), so I was unable to attempt LBC training. I ventured to Smith Rock as planned and set my sights on Rude Boys, one of my dream routes. With no PE training, I was a long shot to send this classic endurance testpiece. I utilized the strategy described above, trying to make long links, and pulling back on quickly after falls in order to cobble together PE "sets." Sure enough, after five of these "training" days spread across a week and a half, I had the endurance to make it through the redpoint crux and send my dream route.

— Mike Anderson

SUMMARY

- A forearm pump is caused by repeated intense contractions that limit blood flow and cause harmful waste to build up.
- PE training causes several enzymatic adaptations that improve the muscles' ability to continue activity beyond the maximum steady state level of effort. Intensity must be high enough to stimulate adaptations in the fast-twitch muscle fibers.
- Adaptations to endurance training are specific, so the training should closely mimic climbing in terms of :
 - Duration
 - Intensity
 - Pace
 - Terrain (hold type, size, wall angle, movements, rests)
- PE training is best performed with Linked Bouldering Circuits (LBC) or Route Intervals, which should be constructed specifically to simulate goal routes and may be designed to lie anywhere on the endurance spectrum.
- As set duration increases, intensity and number of sets performed should decrease. The duty cycle is consistent between PE training exercises, so the longer the set duration, the longer the rest interval.
- Endurance training should progress through a season by increasing the difficulty of moves and/or reducing the rest between sets.
- Skill-development activities should be incorporated into PE workouts.
- PE workouts occur just before and during the Performance Phase, and should be followed by 48 to 72 hours of rest.